

1. Administrative & Study Identification

Full Study Title: A Study of MK-8527 in Healthy Lactating Female Participants (MK-8527-009) **Short Title/Acronym:** MK-8527 PK in Lactating Females **Protocol Number:** MK-8527-009 **NCT Number:** NCT06580587 **Sponsor Name:** Merck Sharp & Dohme LLC **Investigational Product:** MK-8527 (Nucleoside Reverse Transcriptase Translocation Inhibitor / NRTTI) **Phase of Development:** Phase I **Medical Monitor:** Merck Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective **Primary Objective:** To evaluate drug concentrations of MK-8527 in breast milk following a single oral dose in healthy lactating female participants. **Secondary Objectives:** To evaluate the pharmacokinetic (PK) profile of MK-8527, including maximum observed concentration (\$Cmax\$), time to \$Cmax\$ (\$Tmax\$), and area under the curve (\$AUC0-120hrs\$ and \$AUC0-inf\$) in blood and plasma.

2.2 Background & Rationale MK-8527 is an investigational once-monthly oral NRTTI in development for HIV pre-exposure prophylaxis (PrEP). To ensure the safety of this therapeutic approach in all populations, it is critical to learn how MK-8527 moves through a healthy person's body over time and specifically to determine the amount of the drug transferred into breast milk that a nursing infant could potentially receive.

3. Study Design & Methodology

3.1 Design Framework **Study Type:** Interventional **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration **Target \$N\$:** Not publicly specified (Standard Phase 1 sample size) **Number of Sites:** Single/Multi-center Phase 1 Unit(s) **Estimated Study Start:** 2024 **Estimated Primary Completion:** Ongoing/Recruiting

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Female, \$\ge 18\$ years of age. 2. Is at least 6 weeks postpartum at the time of administration of study intervention, following the delivery of a healthy singleton neonate. 3. Is willing and able to express breast milk at least twice daily for at least 120 hours after enrollment. 4. Is

willing to discontinue breastfeeding from the time of administration of study intervention until at least 6 weeks following the administration of study intervention (this includes avoiding feeding expressed milk to the infant).

4.2 Exclusion Criteria 1. Medical conditions or history that could interfere with the study evaluations or pose a safety risk. 2. Delivery of multiple birth neonates (e.g., twins). 3. Concurrent use of investigational therapies or medications known to interact with MK-8527 absorption, distribution, metabolism, or excretion.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Single dose **Route of Administration:** Oral **Duration of Treatment:** Single administration followed by 120 hours of intensive PK and breast milk expression.

5.2 Concomitant & Prohibited Therapy Permitted: Standard allowed medications as defined by the Phase 1 clinical unit. **Prohibited:** Use of any investigational agents or medications known to affect the pharmacokinetics of MK-8527 prior to and during the study.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to -1	Informed Consent, Medical History, Physical Exam, Eligibility Labs
Baseline	Day 0	Confinement, First Dose of MK-8527, Intensive PK Blood/Plasma Sampling
Interim	Day 1 to Day 5	Expression of breast milk 2 times daily for 120 hours, serial blood/plasma sampling
End of Study	Week 6	Final Safety Assessment, confirmation of breastfeeding discontinuation period completion

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Amount of MK-8527 excreted in breast milk (0-120hrs) after the administration of a single oral dose. **Measurement Method:** Central Laboratory Analysis of expressed breast milk.

7.2 Secondary & Exploratory Endpoints 1. AUC_{0-120hrs} and AUC_{0-inf} of MK-8527 in blood/plasma. 2. C_{max} and T_{max} of MK-8527 in plasma. 3. Ratio of MK-8527 in breast milk to blood/plasma used to estimate the potential dose of MK-8527 an infant would receive.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring during clinical confinement and follow-up via HCP interview and laboratory assessments. **Serious Adverse Events (SAEs):** Reported to the Sponsor and IRB/IEC within 24 hours of site awareness. **Data Monitoring Committee (DMC):** Internal safety review committee.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Pharmacokinetic (PK) Analysis Set (all subjects who received the study dose and have sufficient concentration data). **Sample Size Justification:** A standard Phase 1 sample size is utilized to adequately characterize the PK parameters and estimate the milk-to-plasma ratio without formal hypothesis testing. **Handling of Missing Data:** PK parameters calculated using standard non-compartmental analysis; missing data points are typically excluded or imputed following standard PK guidelines.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Continuous on-site monitoring during confinement followed by remote/on-site data verification. **Audit Trail:** Detailed electronic Case Report Form (eCRF) completion, maintaining strict logs for milk expression and PK sampling times.

11. Study Identifiers & Governance

Primary ID: MK-8527-009 **Registry Number:** NCT06580587 **Sponsor/Lead Organization:** Merck Sharp & Dohme LLC **Study Title:** A Study of MK-8527 in Healthy Lactating Female Participants (MK-8527-009) **Official Regulatory Title:** A Study of MK-8527 in Healthy Lactating Female Participants

12. Clinical Framework (PK & Lactation Specific)

Primary Condition: Healthy Lactating Females **Diagnosis Threshold:** ≥ 6 weeks postpartum with a healthy singleton neonate **Medical Methodology:** Pharmacokinetic (PK) Evaluation **Optimization Target:** MK-8527 PK profile characterization and infant exposure estimation via breast milk

13. Study Procedures & Methodology

Management Pathway: Strict inpatient clinical unit pathway for PK sampling **Education Protocol (HCP):** Protocol-specific training for accurate timing of blood draws and milk processing **Education Protocol (Patient):** Onsite training on breast pump utilization and strict instructions on the mandatory 6-week cessation of breastfeeding **Quality Audit Mechanism:** Adherence monitoring of the 120-hour expression timeline and continuous AE tracking

14. Observation & Follow-Up Schedule

Total Duration: 6 weeks post-dose **Onsite Visit Cadence:** Confinement up to 120 hours (Days 1-5) post-dose **Remote Monitoring Frequency:** Intermittent follow-up calls up to Week 6 **Checklist Review Interval:** Final evaluation at the end of the 6-week safety and breastfeeding cessation period

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---	
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]					